

WEIR PROTOCOL

Version: 1.0 (Production)

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Sponsor: Creditcoin & Credit Labs

Network: Creditcoin CC3 & Ethereum

Sepolia

TECHNICAL
ARCHITECTURE

WHITEPAPER

&

SYSTEM

Abstract: Real-World Asset (RWA) tokenization suffers from an institutional counterparty trust gap: while commercial payments occur in liquid fiat or stablecoins on external chains, fractional cap table accounting and dividend distributions settle on high-throughput execution layers. Current bridges rely on centralized multisigs and custodial keepers that reintroduce skimming, delays, and insolvency risk. **WEIR** (/wiər/) introduces an autonomous, oracle-free revenue distribution protocol anchored by Creditcoin's Universal Smart Contracts (USC / Attestcoin Protocol). By verifying Ethereum deposit receipts directly inside runtime precompiles (0x00 ... 0FD2) and executing an $O(1)$ scalable dividend indexing algorithm, WEIR creates an unskimmable, verifiable corporate dividend pipe.

Track: RWA &
DeFi

EVM Chain: 102031
(CC3)

Attestcoin Precompile:
0x0FD2

Complexity: $O(1)$
Scalable

1. Introduction & The Middleman Problem

In asset tokenization (commercial solar farms, equipment leasing, real estate, and delivery fleets), operating assets produce tangible revenue in the real world. However, the transmission of that revenue to on-chain investors is opaque. Sponsors collect lease payments into traditional off-chain accounts and distribute fractional payouts manually, often presenting reports with opaque "operating deductions."

Investors have no cryptographic certainty that gross revenue receipts match real cash flows. Centralized bridge oracles and multisigs fail to provide sovereign security guarantees. WEIR eliminates this discretion by transforming commercial cash flows into a deterministic mathematical distribution pipe.

2. System Architecture

WEIR uses a dual-contract architecture connecting Ethereum Sepolia (liquidity layer) with Creditcoin CC3 Testnet (settlement layer):

```
[ Ethereum Sepolia (Chain 11155111) ]
└─ WeirVault.sol (0x13C40f20908C66A9c31D6102234c1095E12A31e3)
    |
    └─ Emits: RevenueDeposited(assetId, grossAmount, period, payor)
        |
[ Attestcoin Consensus & Proof Generation ]
└─ Validator Attestation: Ethereum block finalized on Creditcoin
└─ Proof Builder Service: Generates Merkle receipt & continuity proofs
    |
[ Creditcoin CC3 Testnet (Chain 102031) ]
└─ WeirDistributionASC.sol (0xe01236C5Fd875b47A8e6DE4F5c4B39959bba8d3C)
    └─ 1. Interfaces with Block Prover Precompile (0x0FD2)
    └─ 2. Decodes receipt bytes via on-chain EvmV1Decoder
    └─ 3. Enforces strict replay protection (processedQueries)
    └─ 4. Settles 0(1) cumulative dividend per share
```

3. Attestcoin Protocol / USC Integration

The Attestcoin Protocol (Universal Smart Contracts) provides native decentralized infrastructure for verified cross-chain data. Rather than trusting off-chain nodes to sign attestations, Creditcoin validators verify Ethereum block headers directly in consensus.

WEIR executes proof verification through Creditcoin's native Block Prover Precompile located at 0x00FD2:

```
// WeirDistributionASC.sol
function verifyAndDistribute(
    uint64 chainKey,
    uint64 headerNumber,
    bytes calldata txBytes,
    Proof calldata proof
) external nonReentrant {
    bytes32 txKey = keccak256(abi.encodePacked(chainKey, headerNumber, txBytes));
    require(!processedQueries[txKey], "Query already processed");

    // Native Precompile Verification
    bool valid = VERIFIER.verifyAndEmit(chainKey, headerNumber, txBytes, proof);
    require(valid, "Attestcoin precompile verification failed");
    processedQueries[txKey] = true;

    // Unpack receipt and update dividend index
    ...
}
```

4. The O(1) Scalability Invariant

Traditional dividend distributions loop over investor arrays—a design pattern that inevitably exceeds block gas limits when investor counts exceed a few dozen. WEIR implements the MasterChef / Synthetix cumulative dividend index model:

$$\Delta \text{ Index} = (\text{Gross Revenue Deposited} \times 10^{18}) / \text{Total Outstanding Shares}$$

When revenue arrives, the contract executes a single storage write regardless of whether the asset has 3 or 100,000 fractional investors. Individual investors claim their accrued yield on-demand using a pull-claim model:

$$\text{Owed Yield} = \text{InvestorShares} \times (\text{CurrentIndex} - \text{UserLastClaimedIndex}) / 10^{18}$$

5. Live Testnet Deployment & Empirical Evidence

The entire WEIR system is fully deployed, containerized, and verified across production testnets:

Component	Network	Address / Identifier
WeirVault	Ethereum Sepolia	0x13C40f20908C66A9c31D6102234c1095E12A31e3
WeirDistributionASC	Creditcoin CC3	0xe01236C5Fd875b47A8e6DE4F5c4B39959bba8d3C
Native Verifier	Creditcoin CC3	0x00FD2
Relayer Daemon	Railway Worker	Deployment 518b3df0-6cff-463b-8c88-dc0ddeb34831
Web Console	Vercel Production	https://weir-protocol.vercel.app/console

Verified Production Run: An end-to-end live deposit of 500 USDC on Sepolia (Tx: 0xc240fa864e5d906465406d6e72c5a197ba722557d9845b043ea7224a275d6af4) was autonomously relayed by Railway, verified via Precompile 0x0FD2, and settled on Creditcoin CC3 (Tx: 0x789fb1e56ad015631fd3dd3519d8d89dc1dc005d6af9fdf2380353c84f426b66), advancing the dividend index from 1,050,000 to 1,100,000.

6. Conclusion & CEIP Acceleration

WEIR provides the missing trustless financial plumbing for institutional RWA on Creditcoin. By utilizing Attestcoin Protocol for unforgeable cross-chain receipts and Creditcoin for near-zero gas dividend distribution, WEIR is primed for acceleration under the Creditcoin Ecosystem Investment Program (CEIP).